

TOTAL CLASSES 25 Weekend Classes 50 Hands-On Lab Hours	TAKE-HOME KITS 10 Individual Builds 1 per student to keep	STUDIO GROUP LABS 10 Reusable Stations Practiko Master Bundles	MILESTONE MILESTONES 4 Challenges + Expo ₹0 Hardware Cost (Build Reuse)
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 **Milestone Evaluation & Zero Hardware Cost Policy:** Classes 6, 12, 18, and 24 are gamified milestone trials. Students bring and reuse the robotic mechanisms, gliders, circuits, and sensors built in the preceding 5 classes. Zero new kit purchase is required for assessment classes, maximizing hands-on testing and iterative engineering.

Junior Batch — Complete 25-Class Syllabus (Ages 6–9 / Grades 1–4)

Foundational STEM • Mechanics • Aeromodelling • Block Code

Cls	Ph	Curriculum Module	Format	Assigned Kit / Activity Name	Vendor / SKU	Core Learning Outcome & Deliverable
01	P1	Module 1: Meet the Robots	 Group Lab #1	Human vs. Robot Explorer Station	Practiko PRA-MECH-01	Tools vs. machines vs. autonomous robots; sensory organs vs. robot sensors.
02	P1	Module 2: Instructions to Robots	 Group Lab #2	Floor Grid Robot Maze & Code Cards	Good Soil STUDIO-CODE-01	Unplugged step sequencing, directional commands, path debugging, algorithmic logic.
03	P1	Module 3: Robot Components	 Group Lab #3	5 Organs of a Robot Dissection Lab	Practiko PRA-CIRC-01	Power (Battery), Input (Sensors), Brain (Controller), Output (Motors), Chassis.
04	P1	Module 4: BreadBoard Intro	 Giveaway #1	Junior Snap / Pinboard LED Circuit	Good Soil STUDIO-SNAP-01	Solderless connections, power rails, closed loops, LED lighting without soldering.
05	P1	Module 5: Powering a Robot	 Giveaway #2	Electric Propeller & Aerodynamic Wind Car	Pludo (Robocraze) WEB-PLU-015	DC motor rotation polarity, battery circuits, aerodynamic thrust and propulsion.
06	P1	 Milestone Assessment 1	 Milestone Test	Island Escape Challenge (Reuses Class 1–5 Builds)  100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-01	Milestone Evaluation: Students bring catapult and propeller car builds from Classes 1–5 to launch SOS signals and negotiate obstacle mazes. (Badge: <i>Junior Explorer</i>)
07	P2	Module 6: Sensors — Part I	 Giveaway #3	Wooden Glow Table Lamp & Switch Circuit	Kiddale WEB-KDA-042	Light-dependent resistors (LDR), automatic dark-sensing switch circuits.

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08	P2	Module 7: Sensors & Buzzers	 Giveaway #4	Morse Code Telegraph & Buzzer Station	Good Soil STUDIO-ELEC-01	Piezo buzzers, sound oscillation, binary sound signals, Morse code signaling.
09	P2	Module 8: Sensor & Switches	 Group Lab #4	2-Way Reversible Motor & Railway Gate	Practiko PRA-MECH-02	Push buttons, toggle switches, polarity reversal, forward/reverse motor motion control.
10	P2	Module 9: Motors & Movement	 Group Lab #5	Mechanical Gearbox & Carousel Machine	Kiddale WEB-KDA-032	Spur gears, driver vs. driven gears, speed vs. torque, gear ratio power transmission.
11	P2	Module 10: Build a Moving Bot	 Giveaway #5	Junior 2-Wheel Obstacle Rover Build	Kiddale WEB-KDA-140	Chassis construction, dual DC motors, battery wiring, directional alignment.
12	P2	 Milestone Assessment 2	 Milestone Test	STEAM Carnival Derby (Reuses Class 7–11 Builds)  100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-02	Milestone Evaluation: Teams combine paddle steamer boats, gear carousels, and paper circuit signaling into an automated derby arena. (Badge: <i>Master Crafter</i>)
13	P3	Module 11: Microcontrollers I	 Group Lab #6	Micro:bit / Block Brain First Blink	Practiko PRA-MCU-01	Embedded microcontroller cores, digital GPIO, block-based icon animation & blink code.
14	P3	Module 12: Microcontrollers II	 Group Lab #7	Interactive Sound & Emotion Display	Practiko PRA-MECH-03	Button triggers, animated face expressions, sound synthesis on OLED displays.
15	P3	Module 13: Microcontroller Project	 Giveaway #6	Smart Security Room Guard / Gate Alarm	Good Soil STUDIO-SEC-01	Ultrasonic sonar distance calibration, automated threshold buzzer alarm triggering.
16	P3	Module 14: Loops & Repetition	 Group Lab #8	Looping Light Patrol & Dancing Bot	Practiko PRA-MECH-04	While/repeat automation loops, continuous routines without human intervention.
17	P3	 Aeromodelling Track I	 Group Lab #9	High-Lift Paper Gliders & Airfoil Launcher	Good Soil CUSTOM-AERO-01	Four forces of flight: Lift, Weight, Thrust, Drag; airfoil camber and wing dihedral.
18	P3	 Milestone Assessment 3	 Milestone Test	Sky Patrol & Beacon Derby (Reuses Class 13–17 Builds)  100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-03	Milestone Evaluation: Target landing accuracy and Morse code communication trial using gliders and buzzer alarm circuits built earlier. (Badge: <i>Flight Cadet</i>)
19	P4	Module 15: Robot Mechanics I	 Giveaway #7	Walking Bionic T-Rex / Crab Linkage	Kiddale WEB-KDA-039	4-bar linkages, crank mechanisms, converting motorized

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						rotation into walking locomotion.
20	P4	Robot Mechanics II	 Giveaway #8	Wooden Catapult & Trajectory Launcher	Practiko WEB-PRA-052	Levers, fulcrums, mechanical advantage, elastic potential energy to kinetic motion.
21	P4	 Aeromodelling Track II	 Giveaway #9	Balsa/Foam Chuck Glider & Flight Tuning	Good Soil CUSTOM-AERO-02	Adjustable elevator trim tabs, ballast clay center of gravity balancing, flight stability.
22	P4	 Aeromodelling Track III	 Group Lab #10	Mini-Drone Flight Discovery Station	BrainHap WEB-BRH-003	Quadcopter aerodynamics: counter-rotating rotors, roll/pitch/yaw joystick flight control.
23	P4	Clean Energy Robotics	 Giveaway #10	Solar-Powered Eco Rover Car	Kiddale WEB-KDA-043	Photovoltaic solar panels, photon-to-electron energy conversion, capstone integration.
24	P4	 Milestone Assessment 4	 Milestone Test	Future City Prototype Arena (Reuses Class 19–23 Builds)  100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-04	Milestone Evaluation: Teams combine walking robots, solar rovers, smart alarms, and drone telemetry into a smart city pitch. (Badge: <i>Young Innovator</i>)
25	P4	 GRAND EXPO	 Public Showcase	Good Soil Junior STEM Expo & Graduation	Good Soil STUDIO-EXPO-01	Robot arena competitions, glider distance trials, parent demos, STEM Passport graduation.

GOOD SOIL STEM LABS — SENIOR SYLLABUS

Target Batch: Senior (Ages 10–14 / Grades 5–9)
Structure: 25 Classes (50 Hours) • 10 Giveaways + 10 Group + 4 Assessments

Advanced Robotics, Embedded C++, Kinematics, Avionics & IoT

Senior Batch — Complete 25-Class Syllabus (Ages 10–14 / Grades 5–9)

Microcontrollers • C++/Logic • Kinematics • Avionics • Cloud IoT

Cls	Ph	Curriculum Module	Format	Assigned Kit / Activity Name	Vendor / SKU	Core Learning Outcome & Deliverable
01	P1	Module 1: Meet the Robots	Group Lab #1	Autonomous Systems & DoF Dissection	Practiko PRA-MECH-S01	Degrees of Freedom (DoF), kinematic chains, Cartesian vs. articulated architectures.
02	P1	Module 2: Instructions to Robots	Group Lab #2	Pseudocode, State Machines & Flowcharts	Practiko PRA-AI-01	Algorithmic decision branches, IF-THEN logic, finite state transition maps.
03	P1	Module 3: Robot Components	Group Lab #3	Hardware BOM & Signal Routing Lab	Practiko PRA-ELEC-S01	Circuit schematics, digital vs. analog signaling, component datasheets, power buses.
04	P1	Module 4: BreadBoard Intro	Giveaway #1	Half-Size Solderless Breadboard Lab	Good Soil STUDIO-BREAD-01	Power rails, 5V/GND bus discipline, IC tie points, jumper wire routing discipline.
05	P1	Module 5: Powering a Robot	Giveaway #2	Multi-Voltage Power Supply & Test Station	Practiko PRA-ELEC-S02	Ohm's Law ($V=IR$), series vs. parallel arrays, multimeter voltage & current draw tests.
06	P1	Milestone Assessment 1	Milestone Test	Circuit Diagnostics Hackathon (Reuses Senior Class 1–5 Builds) ✓ 100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-01S	Milestone Evaluation: Students diagnose injected breadboard faults, calculate resistance, and calibrate power loads. (Badge: <i>Apprentice Engineer</i>)
07	P2	Module 6: Sensors — Part I	Giveaway #3	Precision LDR Light Sensor & Divider Bench	Good Soil STUDIO-SENS-01	Voltage divider formula, analog light curves, pull-up/pull-down threshold tuning.
08	P2	Module 7: Sensors & Buzzers	Giveaway #4	Transistor-Driven Piezo Alarm Circuit	Good Soil STUDIO-SE-02	NPN BJT transistor as a saturation switch, driving high-current buzzers, pulse tones.
09	P2	Module 8: Sensor & Switches	Group Lab #4	Relay Logic & Reversible DPDT Motor Drive	Practiko PRA-MECH-S02	DPDT switch H-bridge emulation, electromechanical relays, machine limit stops.
10	P2	Module 9: Motors & Movement	Group Lab #5	Dual-Motor Gear Ratio & Speed/Torque Bench	Practiko PRA-KIN-S01	Spur and worm gears, gear ratio calculation ($N1/N2$), torque

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						multiplication vs. RPM.
11	P2	Module 10: Build a Moving Bot	Giveaway #5	4WD Heavy-Duty Obstacle Rover Chassis	Good Soil STUDIO-ROV-01	Dual gearmotors, rock-crawling chassis, rocker power switch, obstacle negotiation.
12	P2	Milestone Assessment 2	Milestone Test	Rescue Rover Canyon Race (Reuses Senior Class 7–11 Builds) 100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-02S	Milestone Evaluation: Teams negotiate incline ramps, cable crossings, and cargo retrieval using the 4WD rovers built in Classes 7–11. (Badge: <i>Systems Builder</i>)
13	P3	Module 11: Microcontrollers I	Group Lab #6	Arduino / Microcontroller Embedded Core	Practiko PRA-MCU-S01	ATmega architecture, GPIO registers, C++ firmware upload, serial monitor debugging.
14	P3	Module 12: Microcontrollers II	Group Lab #7	Analog Read & PWM Motor Speed Control	Practiko PRA-MCU-S02	8-bit PWM duty cycle throttling, analogRead() potentiometer/sensor motor curves.
15	P3	Module 13: Microcontroller Project	Giveaway #6	Ultrasonic Radar Distance & Collision Guard	Good Soil STUDIO-RAD-01	HC-SR04 sound speed timing, obstacle detection thresholding, radar sweep routines.
16	P3	Module 14: Loops & Repetition	Group Lab #8	State Machines & Edge-Avoiding Autonomy	Practiko PRA-MECH-S03	Non-blocking timed execution (millis()), finite state machine logic, boundary avoidance.
17	P3	Aeromodelling Track I	Group Lab #9	Rubber-Band Monoplane Endurance Model	Good Soil CUSTOM-AERO-03	Airfoil aerodynamics (camber, chord, AoA), high-aspect-ratio wing glide optimization.
18	P3	Milestone Assessment 3	Milestone Test	Flight Derby & Sensor Arena (Senior Class 13–17 Builds) 100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-03S	Milestone Evaluation: Timed flight endurance trials and acoustic/ultrasonic obstacle avoidance using gliders and radar modules built in Classes 13–17. (Badge: <i>Avionics Specialist</i>)
19	P4	Module 15: Robot Mechanics I	Group Lab #10	Kinematics: Linkages, JCB & Slider-Cranks	Practiko PRA-KIN-S02	Grashof's criteria, slider-crank linkages, JCB excavator arm 3-DoF kinematics.
20	P4	Robot Mechanics II	Giveaway #7	Hydraulic Mech Gripper Robot Arm	Kintaro DIY WEB-KIN-001	Pascal's Law ($P=F/A$), syringe hydraulic actuation, mechanical gripper claw control.
21	P4	Aeromodelling Track II	Giveaway #8	Motorized Electric DIY Aircraft	Pludo (Robocraze) WEB-PLU-074	Thrust-to-weight ratio ($T/W>1$), coreless high-RPM motor, LiPo battery powered flight.

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22	P4	 Aeromodelling Track III	 Group Lab #10	BrainHap Quadcopter Flight & Telemetry	BrainHap WEB-BRH-003	Quadcopter flight controllers, gyro stabilization, PID roll/pitch/yaw joystick control.
23	P4	Advanced IoT Robotics	 Giveaway #10	Smart Connected IoT Rover / Weather Node	Practiko PRA-IOT-01	Wi-Fi telemetry, cloud data streaming, mobile robot remote control capstone.
24	P4	 Milestone Assessment 4	 Milestone Test	Shark Tank & Capstone Defense (Senior Class 19–23 Builds)  100% Kit Reuse (₹0 Cost)	Good Soil STUDIO-ASSESS-04S	Milestone Evaluation: Teams present customized rovers, explain circuit schematics, and defend design iterations before a mentor panel. (Badge: <i>Master Innovator</i>)
25	P4	 GRAND EXPO	 Public Showcase	Good Soil Senior Tech Symposium & Graduation	Good Soil STUDIO-EXPO-02	Autonomous flight trials, robot obstacle runs, tech portfolio defense, medal ceremony.

Milestone Assessment 100-Point Scoring Rubric & Methodology

Non-Paper Evaluation • Practical Competency

1. Engineering Build (30 Pts)

Assembly precision, solderless breadboard discipline, smooth gear mesh, zero short-circuits.

2. First-Principles (25 Pts)

Student verbally explains the underlying physics, current loops, gear ratio math, or flight forces.

3. Troubleshooting (20 Pts)

Methodical fault tracing: multimeter testing, loose jumper identification, trim adjustment.

4. STEAM Extension (15 Pts)

Creative mechanism enhancements, custom chassis tailoring, aerodynamic optimization.

5. Presentation (10 Pts)

Clear articulation of technical decisions, teamwork, peer collaboration, and defense.